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A CASE STUDY REPORT

ON

CMS PLATFORMS USED IN NEPAL

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A CASE STUDY REPORT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR IN SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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Executive Summary

A content management system (CMS) is the software layer that allows individuals and organisations to create, edit, organise, and publish digital content on the web without needing to write code for every page. It sits between the raw storage of content, text, images, video, and structured data, and the presentation layer through which that content ultimately reaches an audience, whether via a browser, a mobile application, or an application programming interface (API) consumed by a third-party channel. This report presents a comprehensive, technically grounded case study of the CMS landscape, examining the architecture and comparative strengths of major categories of platform, and narrowing its focus to a detailed analysis of how these categories are actually deployed across Nepal's news media, small and medium enterprises (SMEs), e-commerce sector, and government institutions.

The global CMS market spans a spectrum from free, community-maintained open-source projects such as WordPress, Joomla, and Drupal, through proprietary software-as-a-service (SaaS) website builders such as Wix, Squarespace, and Shopify, to the newer category of headless or API-first content platforms such as Contentful, Strapi, and Sanity. Each category represents a distinct trade-off between cost, control, technical skill requirement, and long-term scalability, and organisations frequently combine several categories simultaneously as part of a broader digital strategy.

Within Nepal, publicly available web-technology tracking data indicates that WordPress is, by a wide margin, the most widely detected CMS among Nepali websites, followed at a considerable distance by Blogger and Drupal, with smaller shares held by Joomla, Squarespace, Webflow, and an emerging cluster of headless platforms. This concentration reflects Nepal's cost-sensitive SME sector, a mature local freelance and agency ecosystem built around WordPress, and the relatively low barrier to entry that open-source software provides in a market where formal information-technology budgets remain modest for the majority of businesses.

The report also documents a clear bifurcation in Nepal's digital ecosystem: while WordPress and similar off-the-shelf systems dominate the small-business, blogging, and news-portal segments, the country's largest media houses and e-commerce platforms increasingly rely on custom-built, cloud-native, or headless content systems to meet performance, scalability, and multi-channel publishing needs. Government institutions, meanwhile, depend on a hybrid

model that combines centrally coordinated infrastructure under the National Information Technology Center (NITC) with vendor-built platforms of varying technical maturity, producing an uneven but gradually modernising public-sector web presence.

Key recommendations arising from the study include continued investment in local web-development capacity and Nepali-language and Bikram Sambat-aware plugins and themes; wider adoption of security best practices, automated backups, and update discipline among small-business website operators; greater standardisation of government web platforms under unified technical governance; and a gradual, capacity-appropriate exploration of headless and API-first architectures by organisations whose content needs to reach multiple channels such as mobile applications, digital signage, and social platforms. Collectively, these findings and recommendations are intended to serve as a practical reference for students, web professionals, business owners, and policymakers seeking to understand the trade-offs between CMS categories and their real-world implications for Nepal's growing digital economy.

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CHAPTER 1: INTRODUCTION

A content management system is, at its core, an application, usually accessed through a web browser, that separates the creation and editing of content from the underlying code, data storage, and visual design of a website. It provides a dashboard through which non-technical users can add text, images, video, and other media, while the CMS itself handles how that content is stored in a database, templated into presentable pages, cached for performance, and served to visitors across desktop and mobile devices. In doing so, a CMS abstracts away the complexity of raw web development, HTML markup, server configuration, and database queries, and replaces it with an editorial workflow that a marketing team, a journalist, or a small-business owner can operate directly.

Over the past two decades, the falling cost of hosting, the maturity of open-source ecosystems, and the explosive growth of e-commerce and digital media have made CMS platforms central to how organisations of every size establish a presence online. What began as simple blogging tools in the early 2000s has evolved into a rich, multi-billion-dollar software category encompassing enterprise digital-experience platforms, no-code website builders, and API-first content infrastructure that feeds websites, mobile apps, kiosks, and even Internet-of-Things displays from a single content repository.

1.1 Theoretical Background

Conceptually, every CMS performs four core functions regardless of its category: content creation (authoring tools, rich-text or block editors, media libraries), content management (versioning, workflow states such as draft and published, user roles and permissions), content storage (a database or content repository, typically relational such as MySQL or PostgreSQL, though some modern systems use document-oriented or graph-based storage), and content delivery (the mechanism by which stored content is rendered or transmitted to an end-user channel). The way a given platform implements the fourth function, delivery, is what principally distinguishes the major architectural categories examined in this report: a traditional or “coupled” CMS renders finished HTML pages directly, whereas a headless or “decoupled” CMS exposes content purely through an API and leaves rendering to a separate front-end application.

This theoretical distinction has practical consequences for cost, technical skill requirement, and scalability, all of which are explored in depth later in this report. It also has consequences for how content strategy is designed: an organization publishing to a single website has different needs from one that must simultaneously serve a website, a native mobile application, and a set of social-media integrations, and the choice of CMS architecture materially affects how easily that multi-channel requirement can be met.

1.2 Scope of the Report

The scope of this report is limited to web content management systems used for publishing websites, blogs, and online storefronts. It does not extend to enterprise document-management systems, learning-management systems, or digital-asset-management platforms used purely for internal organizational purposes, except where relevant to illustrate the broader CMS landscape. Geographically, the report examines global platforms as a point of technical reference before narrowing to a detailed sector-by-sector case study of Nepal, covering small and medium enterprises, independent and mainstream news media, e-commerce, and government institutions. The report does not attempt an exhaustive market-share census; rather, it uses publicly available web-technology tracking data, general industry knowledge, and structured comparative analysis to describe general patterns of adoption that a student, practitioner, or policymaker can reasonably rely upon.

1.3 Methodology

This report is a desk-based case study, compiled from publicly available information, including CMS vendor documentation, web-technology tracking services such as Built With and W3Techs, industry reporting on the global CMS market, and general knowledge of the Nepali web-development sector accumulated through community events, agency portfolios, and published commentary. Comparative tables are used throughout to summaries the characteristics of each platform category on a consistent basis across five evaluation criteria: cost, ease of use, customizability, security, and performance. Because CMS market share and platform capabilities evolve rapidly, new versions are released frequently, and vendor pricing changes without notice, figures relating to adoption and pricing in this report should be treated as indicative of the general

pattern observed at the time of writing rather than as real-time data, and readers requiring current figures are encouraged to consult live tracking services directly.

The comparative framework applied in Chapter 6 follows a structured scoring logic: each CMS category is assessed against the five criteria using a combination of qualitative description and, where meaningful, indicative cost ranges converted to Nepali rupees at approximate prevailing exchange rates for illustrative purposes. This approach mirrors the kind of lightweight, criteria-based evaluation that a small organization or a student project might realistically perform when selecting a platform, rather than a formal total-cost-of-ownership audit of the kind an enterprise procurement team would commission.

1.4 Research Questions / Problem Definition

The report is structured around the following guiding research questions:

- What are the principal architectural and business-model categories into which CMS platforms can be classified, and what trade-offs does each category impose?
- How are these categories of platform actually adopted across different segments of Nepal's web ecosystem, small businesses, news media, e-commerce, and government?
- What local factors, cost sensitivity, payment infrastructure, logistics, language and calendar requirements, and developer availability, shape CMS adoption patterns in Nepal specifically?
- How do the major CMS categories compare when evaluated systematically on cost, ease of use, customisability, security, and performance?
- What risks and operational challenges, both global and Nepal-specific, affect organisations relying on each category of platform, and how can these risks be mitigated?
- What emerging trends are likely to shape the future of CMS adoption globally and within Nepal over the coming years?

The underlying problem this report addresses is one of informed choice: organisations in Nepal, ranging from a single-person freelance business to a large national ministry, frequently select a CMS platform based on cost or on the recommendation of a single vendor, without a structured

understanding of the longer-term trade-offs involved. By presenting a systematic classification, a sector-specific case study, and a comparative framework, this report aims to equip such organizations, and the students who will go on to advise or build for them, with a clearer basis for decision-making.

The remainder of this report proceeds as follows. Chapter 2 establishes the architectural and categorical classification used throughout the analysis. Chapter 3 sets out the study's objectives in itemised form. Chapter 4 examines the major global CMS platforms in technical depth, before Chapter 5 narrows the focus to a sector-by-sector case study of Nepal covering small businesses, media, e-commerce, and government. Chapter 6 presents a structured comparative analysis across five evaluation criteria, and Chapter 7 discusses the risks and operational challenges facing both global and Nepal-specific deployments. Chapter 8 looks ahead to emerging trends, and the report closes with a conclusion and a set of stakeholder-specific recommendations in Chapters 9 and 10.

CHAPTER 2: ARCHITECTURAL AND CATEGORICAL CLASSIFICATION OF CMS PLATFORMS

Before examining individual platforms, it is useful to classify CMS software along several independent dimensions. This classification provides the analytical framework used throughout the comparative analysis presented later in this report, and it clarifies that the categories are not mutually exclusive: a single organisation's web presence typically combines several of these models simultaneously, for example running a self-hosted WordPress blog alongside a SaaS e-commerce storefront and a headless CMS feeding content to a mobile application.

Classification Basis	Categories	Representative Examples
Hosting model	Self-hosted / Software-as-a-Service (SaaS)	WordPress.org (self-hosted); Wix, Shopify (SaaS)
Architecture	Traditional (coupled) / Headless (API-first)	WordPress, Joomla (coupled); Contentful, Strapi (headless)
Licensing	Open-source / Proprietary	WordPress, Drupal, Joomla (open-source); Wix, Squarespace (proprietary)
Primary use case	Blogging & general websites / E-commerce / Enterprise	WordPress (general); Shopify, WooCommerce (e-commerce); Drupal (enterprise)
Rendering strategy	Server-side rendering / Static site generation / Client-side rendering	WordPress (server-side); Gatsby with a headless CMS (static); Strapi + React SPA (client-side)

Table 1: Classification Framework of CMS Platforms

2.1 Coupled (Traditional) versus Headless (API-First) Architecture

A coupled, or traditional, CMS manages content and renders the final web page within the same system: when a visitor requests a page, the CMS retrieves the relevant content from its database, inserts it into a template, and returns a complete HTML document to the browser. This is the model used by WordPress, Joomla, and Drupal in their default configurations, and it has the advantage of simplicity, a single system to install, configure, and maintain, with content editors able to preview exactly what a visitor will see.

A headless CMS, by contrast, separates content management entirely from content presentation. The CMS exposes content through a REST or GraphQL API, and a separate front-end application,

which may be a website built with a JavaScript framework such as React or Vue, a native mobile application, or an entirely different channel such as a digital kiosk, is responsible for requesting that content and rendering it. This decoupling allows the same content to be published once and consumed by multiple channels simultaneously, at the cost of requiring a dedicated front-end development effort that a coupled CMS does not.

A middle ground, sometimes called a “hybrid” or “decoupled” CMS, retains a traditional content-editing interface and default front-end rendering while also exposing an API that additional channels can consume, WordPress, for instance, ships with a built-in REST API that allows it to function in either a coupled or a hybrid capacity depending on how it is deployed.

2.2 Open-Source versus Proprietary Software-as-a-Service

Open-source CMS software, such as WordPress.org, Joomla, and Drupal, is distributed under licences that allow anyone to install, modify, and redistribute the underlying code without a licensing fee. The organisation deploying the software is responsible for arranging its own hosting, applying security updates, and managing backups, but in exchange gains full control over the codebase, the ability to modify functionality directly, and freedom from vendor lock-in.

Proprietary SaaS platforms, such as Wix, Squarespace, and Shopify, bundle software, hosting, and ongoing maintenance into a single subscription product operated entirely by the vendor. The customer never touches server infrastructure or source code; instead, they interact with the platform through a web-based dashboard and, in most cases, a visual drag-and-drop page builder. This model minimises technical barriers to entry but constrains customisation to whatever options the vendor has chosen to expose, and it ties the customer's content and configuration to that vendor's ecosystem for as long as the site remains live.

2.3 Hosting Models

Beyond the open-source/proprietary distinction, hosting arrangements for open-source CMS software themselves vary considerably in a way that materially affects cost, performance, and maintenance burden. Shared hosting, in which many websites run on a single physical server and share its resources, is the cheapest and most common arrangement for small Nepali websites, typically costing the equivalent of a few hundred to a few thousand Nepali rupees per month, but

it offers limited performance headroom and weaker isolation from other tenants on the same server. Virtual private server (VPS) and dedicated hosting provide greater control, predictable performance, and the ability to install custom software, at correspondingly higher cost and a greater need for server-administration expertise. Managed WordPress hosting, offered by specialist providers, bundles automatic updates, backups, and performance optimisation into a premium hosting product specifically tuned for one CMS. Finally, cloud and content-delivery-network (CDN) based hosting, increasingly used by Nepal's larger media houses, distributes content across globally or regionally located edge servers to reduce latency for geographically dispersed audiences and to absorb traffic spikes that a single shared-hosting server could not withstand.

2.4 Composable and Hybrid Architectures

A more recent evolution in enterprise content management is the “composable” or MACH (Microservices, API-first, Cloud-native, Headless) architecture, in which an organisation assembles its digital platform from multiple best-of-breed services, a headless CMS for content, a separate commerce engine for transactions, a dedicated search service, and a customer-data platform for personalisation, connected through APIs rather than relying on a single monolithic system to perform every function. While composable architecture remains rare among Nepali organisations at present, given the specialised engineering capacity it requires, it represents the direction in which the largest global digital-experience deployments are moving, and it is a useful reference point for understanding why Nepal's largest media and e-commerce platforms are beginning to move away from monolithic, off-the-shelf CMS software even before adopting a fully composable stack.

Taken together, these four classification dimensions, hosting model, architecture, licensing, and rendering strategy, are not independent choices made once and fixed forever; rather, they represent points along which an organisation's digital infrastructure typically evolves as its scale and requirements grow. A typical trajectory observed both globally and within Nepal begins with a self-hosted or SaaS coupled system, migrates toward a hybrid model that exposes an API alongside its default rendering as multi-channel requirements emerge, and, for a small number of the largest organisations, eventually moves toward a fully headless or composable architecture.

Understanding this trajectory is central to the sector-specific findings presented in Chapter 5, where Nepal's news-media and e-commerce organisations are shown to be at varying points along precisely this path.

CHAPTER 3: OBJECTIVES OF THE STUDY

The primary objectives of this study are:

- To examine the structure, technical architecture, and comparative characteristics of major global CMS platforms, including WordPress, Joomla, Drupal, leading SaaS website builders, and headless/API-first systems.
- To examine how these categories of CMS platform are adopted in Nepal, with specific reference to small and medium enterprises, independent and mainstream news media, the e-commerce sector, and government institutions.
- To compare CMS platform categories systematically on the basis of cost (including indicative total cost of ownership over one to five years), ease of use, customisability and extensibility, security and maintenance burden, and performance and scalability.
- To identify the risks and operational challenges facing both global CMS platforms and Nepal's local web-development ecosystem, and to propose practical mitigation strategies for each.
- To assess emerging global and Nepal-specific trends, including headless commerce, edge computing, AI-assisted content tooling, and local data-centre expansion, and their likely influence on future CMS adoption.

The secondary objectives of this study are:

- To document the local adaptations that have emerged around global CMS platforms in Nepal, such as Bikram Sambat date-conversion plugins and Devanagari-script publishing tools.
- To catalogue the structure of Nepal's local web-development agency ecosystem and its role in supporting CMS adoption across sectors.
- To provide differentiated, actionable recommendations for four distinct stakeholder groups: small businesses, larger enterprises, government institutions, and educational or training bodies.

CHAPTER 4: GLOBAL LANDSCAPE AND TECHNOLOGY ANALYSIS

Global CMS platforms are the software systems, whether self-hosted or delivered as a service, that power the large majority of the world's websites. They range from free, community-maintained open-source projects to fully managed proprietary platforms, and they underpin everything from personal blogs to large e-commerce operations and enterprise intranets. This chapter examines each major category in technical depth before presenting a benchmark comparison.

4.1 WordPress: Architecture and Ecosystem

WordPress is an open-source content management system originally released in 2003 as a blogging tool and since expanded into a general-purpose website platform. Technically, WordPress is built on PHP and typically backed by a MySQL or MariaDB relational database; content, settings, and metadata are stored across a well-documented database schema of core tables that plugins can extend with their own custom tables or metadata fields. It can be self-hosted using the free WordPress.org software on any compatible hosting environment, or accessed as a managed service through WordPress.com, which trades some configuration flexibility for simplified maintenance.

WordPress is widely reported to power a very large share of all websites on the internet with a CMS detectable by web-technology trackers, supported by an extensive ecosystem of tens of thousands of free and paid themes and plugins that extend its core functionality to e-commerce, membership sites, forums, multilingual publishing, search-engine optimisation, and form handling. Because of its low cost of entry, large community of developers, and extensive plugin ecosystem, WordPress has become the default choice for individuals, small businesses, and many news organisations building a website without a large development budget. The WooCommerce plugin extends WordPress into a capable, fully featured e-commerce platform capable of handling product catalogues, inventory, and payment-gateway integrations, while page-builder plugins such as Elementor and Divi allow non-technical users to design pages visually without writing template code.

WordPress's architecture does carry structural trade-offs. Because plugins can hook into almost any part of the system, poorly coded or abandoned plugins are a leading cause of both security

vulnerabilities and performance degradation, and a heavily plugin-laden WordPress site can become slow and fragile over time without disciplined maintenance. The introduction of the block-based “Gutenberg” editor and, more recently, full-site-editing capabilities has moved WordPress toward a more visual, component-based authoring experience while retaining backward compatibility with its large legacy plugin and theme ecosystem.

4.2 Joomla and Drupal: Structured and Enterprise Content Management

Joomla and Drupal are open-source content management systems that, like WordPress, are self-hosted and free to use, but are generally aimed at users with somewhat greater technical requirements. Joomla, first released in 2005, offers a balance between the ease of use of WordPress and the flexibility of more complex systems, with a built-in, relatively sophisticated access-control system and a content architecture based on customisable article and category structures; it has historically been popular for community, membership, and multilingual institutional websites.

Drupal, first released in 2001, is known for its strength in handling complex, highly structured content types, fine-grained role-based access control, and large-scale enterprise and government websites, at the cost of a steeper learning curve than WordPress. Drupal's “entity” and “field” system allows developers to define arbitrarily complex content structures, for example, a government dataset with dozens of interrelated fields, in a way that WordPress's simpler post-and-page model does not naturally support. This has made Drupal a common choice for large institutional, non-governmental, and public-sector websites internationally, including several national government platforms, where structured data and strict access control outweigh the value of WordPress's simpler authoring experience.

4.3 Proprietary SaaS Website Builders

Proprietary software-as-a-service website builders such as Wix, Squarespace, and Shopify bundle hosting, design tools, and content management into a single subscription product. These platforms are designed to minimise technical barriers, allowing users to build and publish a website through a visual drag-and-drop interface without installing software, configuring servers, or writing code. Shopify in particular has become a dominant platform specifically for online stores, offering integrated payment processing, inventory management, shipping-rate calculation, and a large

marketplace of add-on applications covering everything from email marketing to loyalty programmes.

The convenience of SaaS website builders carries clear trade-offs: users are generally limited to the design and functionality options the platform provides, ongoing subscription fees apply for as long as the site remains online, and moving a website away from the platform to another system, whether to reduce cost or to gain functionality the platform does not offer, can be technically difficult and, in some cases, requires manually rebuilding the site from scratch on the destination system.

4.4 Headless and API-First CMS Platforms

Headless content management systems, such as Contentful, Strapi, and Sanity, separate the storage and management of content entirely from how that content is displayed. Rather than rendering web pages directly, a headless CMS exposes content through an API, typically REST or GraphQL, that can be consumed by a website, a mobile application, a digital signage system, or any other channel simultaneously. Contentful and Sanity are cloud-hosted proprietary services offered on a freemium and usage-based pricing model, while Strapi is open-source and can be self-hosted, giving organisations a choice between managed convenience and full infrastructure control even within the headless category.

This architecture appeals particularly to organisations that need to publish the same content across multiple platforms and that have development teams capable of building custom front-end applications, since the headless CMS itself provides no default visual presentation. It is also well suited to organisations pursuing a “jamstack” approach, in which a static site generator such as Next.js, Gatsby, or Astro pre-builds pages from headless-CMS content at deployment time, yielding very fast page-load performance because no database query is required at the moment a visitor requests a page.

Table 2 below consolidates the preceding discussion of WordPress, Joomla, Drupal, SaaS builders, and headless platforms into a single reference summarising when each was introduced, its primary function, and its typical cost structure, providing a quick-reference companion to the more detailed technical discussion above.

System	Type	Introduced	Primary Function	Typical Cost
WordPress	Open-source, self-hosted	2003	General-purpose website and blog publishing	Free software; hosting and premium plugins extra
Joomla / Drupal	Open-source, self-hosted	2005 / 2001	Community sites (Joomla); complex, enterprise sites (Drupal)	Free software; hosting and development extra
Wix / Squarespace	SaaS website builder	2006 / 2003	Visual, no-code website building	Monthly subscription, typically USD 15–40
Shopify	SaaS e-commerce platform	2006	Online store creation and management	Monthly subscription plus transaction fees
Contentful / Strapi	Headless CMS	2013 / 2015	API-first content delivery to multiple channels	Free tier; usage-based or subscription pricing

Table 2: Summary of Major Global CMS Platforms

4.5 Technical Benchmarks and Architectural Trade-offs

The table below summarises indicative technical characteristics across the major CMS categories, drawing on generally reported performance and architectural patterns rather than a controlled benchmark of specific installations, since real-world performance depends heavily on hosting configuration, caching strategy, and content complexity in every category.

Characteristic	Coupled Open-Source (WordPress/Drupal)	SaaS Builder (Wix/Shopify)	Headless (Contentful/Strapi)
Typical page-render mechanism	Server-side rendering per request, often cached	Server-side rendering managed by vendor	Client-side or build-time rendering via separate front end
Database dependency at render time	Yes, unless page-cached	Abstracted from user	None if statically generated; API call if server-rendered
Multi-channel content reuse	Limited without a REST API layer	Generally single-channel (one website)	Native strength, one content source, many channels
Typical time-to-first-byte under load	Moderate; improved substantially by caching/CDN	Consistently low; vendor-managed infrastructure	Very low for statically generated pages
Extensibility mechanism	Plugins/modules and theme templates	Vendor app marketplace	Custom application code against the API

Table 3: Technical Benchmark Comparison of CMS Architectures

The overarching pattern is that performance and multi-channel flexibility increase as architecture moves from coupled, through SaaS, toward headless and jamstack approaches, but this gain is purchased at the cost of increasing front-end development effort and, in the case of self-hosted

headless deployments, increased infrastructure-management responsibility. This trade-off recurs throughout the sector-specific analysis of Nepal presented in the next chapter, where it explains why small organisations remain concentrated around coupled open-source systems while only the largest, best-resourced organisations move toward headless architectures.

4.6 CMS Adoption in Numbers: A Global Perspective

Independent web-technology tracking services such as W3Techs and BuiltWith consistently report that WordPress is used by a very large proportion of all websites whose content management system can be automatically detected, a figure that has grown steadily over the past decade even as the overall number of distinct CMS products on the market has increased. This dominance is self-reinforcing: a larger installed base attracts more third-party plugin and theme developers, which in turn makes the platform more capable and more attractive to the next wave of adopters, a dynamic economists commonly describe as a network effect operating on software ecosystems rather than on direct user-to-user interaction.

Behind WordPress, Shopify has become the clear leader specifically within hosted e-commerce, reflecting the value merchants place on a single vendor being responsible for payment-compliance certification, uptime, and security patching in a domain where the cost of a data breach or extended outage is directly financial rather than merely reputational. Squarespace and Wix continue to hold a meaningful share of the small-business and portfolio-website segment internationally, generally among users who explicitly prioritise design polish and zero-maintenance operation over deep customisation. Headless platforms such as Contentful, Sanity, and the open-source Strapi remain, in relative terms, a small share of total CMS deployments worldwide, but they are disproportionately represented among large enterprise, media, and multi-brand retail organisations, meaning that their influence on overall content-industry practice is considerably larger than their raw market-share percentage would suggest.

This global pattern, a very large base of small, low-complexity websites concentrated on a small number of accessible open-source or SaaS platforms, alongside a smaller but influential tier of large organisations using custom or headless architectures, is mirrored closely, if at a smaller absolute scale, in the Nepali market examined in the next chapter.

CHAPTER 5: LOCAL SECTOR ANALYSIS: CASE STUDY OF NEPAL'S DIGITAL ECOSYSTEM

Nepal's web ecosystem reflects the same broad categories of CMS platform found globally, but with a distinct pattern of adoption shaped by local cost sensitivity, the availability of Nepali web-development talent, domestic payment and logistics infrastructure, and the specific publishing needs of the country's news media, small-business, and government sectors. This chapter examines each major sector in turn.

5.1 SME and Independent Portal Adoption

Web-technology tracking data compiled by services such as BuiltWith indicates that WordPress accounts for well over half of all content management systems detected among Nepali websites, making it by a substantial margin the most widely used CMS in the country. WordPress is the platform of choice for the majority of Nepali small and medium enterprises, personal blogs, and many independent news portals, largely because it requires no licensing fee, can be deployed cheaply on local or international shared hosting, and is supported by a large pool of local freelance developers and web agencies who are already familiar with its ecosystem.

A dedicated Nepali WordPress community has existed since at least 2012, when the first WordCamp Nepal conference was organised in Lalitpur, and the community has continued to hold annual events bringing together local developers, designers, and site owners to share expertise and showcase locally built themes and plugins. Reflecting the specific needs of Nepali publishers, community-built plugins exist to convert and display dates in the Bikram Sambat calendar alongside or instead of the Gregorian calendar, a requirement common among Nepali news portals, government-adjacent sites, and cultural or event-based businesses that must timestamp content in the local calendar. Similarly, Devanagari-script typography, Unicode font compatibility, and Nepali-language search-engine-optimisation plugins have become a recognisable niche within the local WordPress plugin ecosystem, addressing a gap that mainstream international plugins do not typically prioritise.

Beyond WordPress, Blogger, Google's free, hosted blogging platform, remains the second most detected CMS in Nepal, popular among individual bloggers, students, and smaller organisations

that value its zero-cost hosting and simplicity over the greater flexibility WordPress offers. A smaller but steadily growing group of Nepali freelancers, designers, and small businesses have also begun adopting no-code SaaS website builders such as Squarespace, Webflow, and the built-in site builders offered by hosting providers such as Hostinger, generally for portfolio sites, small service businesses, and clients who explicitly want to avoid ongoing CMS maintenance.

5.2 Enterprise, High-Traffic Media, and Local CDNs

While WordPress dominates the small-site segment, Nepal's larger media houses and e-commerce platforms more frequently rely on custom-built or heavily customised content systems designed for higher traffic volumes and more specific editorial or commercial workflows. Major online news portals with substantial daily readership often move away from off-the-shelf CMS software as they scale, adopting custom publishing systems, dedicated content-delivery infrastructure, or headless architectures that allow the same articles to be published simultaneously to a website, a mobile application, and social-media channels without duplicating editorial effort.

This shift is driven by the practical limitations of shared or even mid-tier VPS hosting under Nepal's variable domestic bandwidth conditions and periods of very high traffic concentration, such as election coverage or breaking-news events, when a standard WordPress deployment on shared hosting can struggle to remain responsive. Larger Nepali media organisations increasingly route traffic through global or regionally present content-delivery networks to cache static assets and, in some cases, full page renders closer to the end user, substantially reducing load times for both domestic and diaspora readers accessing content from outside Nepal. Some larger portals also maintain dedicated in-house technology teams responsible for a custom publishing back end, decoupled from any single off-the-shelf CMS, giving them full control over caching strategy, advertising-technology integration, and multi-channel distribution.

Nepal's leading e-commerce platforms typically operate on custom or heavily modified e-commerce back ends rather than generic storefront builders, reflecting the need to integrate with local payment systems, domestic logistics and cash-on-delivery processes, and inventory or catalogue structures that differ from the assumptions built into many international e-commerce platforms designed primarily around card-based payment and courier-based fulfilment models common in North America and Europe.

5.3 Payment Gateway Integration and Local Logistics

A defining characteristic of e-commerce CMS deployment in Nepal is the need to integrate with domestic digital-payment infrastructure, principally connectIPS, eSewa, and Khalti, alongside continued heavy reliance on cash-on-delivery as a fulfilment and payment model for a large share of online orders. International e-commerce platforms and generic WooCommerce payment plugins generally do not support these Nepali gateways out of the box, so integration is typically achieved either through community-built or agency-developed plugins that connect WooCommerce or a custom cart system to the relevant gateway's API, or through fully custom-built checkout flows on bespoke e-commerce back ends used by larger retailers.

CMS Category	Local Payment Integration Approach	Typical Adopter
WordPress / WooCommerce	Community or agency-built plugins connecting to eSewa, Khalti, or connectIPS APIs	Small and medium online retailers
SaaS builders (Shopify, Wix)	Limited native support; often requires custom app development or manual workarounds	Retailers targeting international as well as domestic customers
Custom-built back ends	Direct API integration engineered in-house against Nepali gateway documentation	Nepal's largest e-commerce platforms
Cash-on-delivery workflows	Order-status and logistics-tracking fields added via plugins or custom database fields	Majority of Nepali online retailers regardless of CMS

Table 5: Local Payment Gateway Integration Approaches by CMS Category

Logistics integration presents a parallel challenge: Nepal's courier and last-mile delivery landscape is fragmented relative to markets with a small number of dominant national carriers, and many online retailers rely on a combination of their own delivery staff for in-valley orders and third-party courier services for orders outside the Kathmandu valley. CMS and e-commerce platforms used in Nepal must therefore often be customised to support multiple, sometimes manually reconciled, fulfilment workflows rather than a single automated shipping-API integration, a pattern that further explains the local preference for custom or heavily modified e-commerce back ends among higher-volume retailers.

5.4 Government of Nepal Web Infrastructure

Government of Nepal websites are coordinated in part through the National Information Technology Center (NITC), established in 2001 as the government's central information-technology implementing body, which hosts and supports a range of government sites and

applications and provides consultancy and training to government offices. NITC's role includes maintaining shared hosting infrastructure and, in some cases, common design and branding standards intended to give government websites a more consistent appearance and technical baseline.

In practice, however, individual ministries, departments, and local governments have historically procured their websites from a range of private vendors operating with varying degrees of technical sophistication, resulting in a government web presence that combines centrally supported infrastructure with independently developed sites of varying technical consistency and CMS maturity. Some ministerial and departmental websites are built on modern, actively maintained CMS installations with reasonable security practices, while others remain on outdated software versions, reflecting the absence of a single enforced technical standard across all levels of government. This pattern mirrors a broader trend seen in other developing digital-government contexts internationally, where central information-technology bodies promote shared standards and hosting while day-to-day website development and content publishing remains distributed across many individual government offices with differing budgets and technical capacity.

Localisation requirements specific to Nepal's government sector include Devanagari-script publishing, Bikram Sambat date display for notices and circulars, and, in many cases, bilingual Nepali-English content structures, each of which has driven the same kind of locally built plugin and template ecosystem observed in the news-media and SME sectors, albeit implemented with less consistency across the government sector as a whole given its fragmented procurement model.

5.5 Local Agency Ecosystem and Development Community

Nepal's CMS ecosystem is supported by a layered development community. At the base is a large and active pool of freelance WordPress developers who build and maintain small-business and blog sites, sustained by a community that has been organising around WordCamp Nepal since 2012. Above this sit established web-development agencies of varying size, ranging from small two- or three-person studios to larger, more formally structured IT firms, which build custom sites, e-commerce platforms, and government vendor sites, typically for clients whose requirements exceed what an off-the-shelf WordPress installation can comfortably support. Finally, in-house media-technology teams within Nepal's largest news organisations maintain custom publishing

platforms, focusing specifically on performance and multi-channel publishing needs that a generic CMS vendor relationship would not adequately serve.

Category	Typical Role	Notable Feature
Freelance WordPress developers	Build and maintain small-business and blog sites	Large, active community since WordCamp Nepal (2012)
Web development agencies	Build custom sites, e-commerce platforms, and government vendor sites	Range from small studios to established IT firms
In-house media technology teams	Maintain custom publishing platforms for major news portals	Focus on performance and multi-channel publishing
NITC	Central government IT coordination	Hosting, consultancy, and training for government offices

Table 6: Notable Nepali Web Development Agencies and Roles

Taken together, these characteristics indicate an ecosystem with a strong concentration around a small number of dominant platforms, chiefly WordPress, reflecting cost sensitivity and the availability of local development skills; a clear divide between small, off-the-shelf CMS deployments and the custom or headless systems used by larger media and e-commerce operators; meaningful local adaptation of global platforms to meet Nepal-specific publishing requirements; government web infrastructure that combines centrally coordinated hosting with development still substantially outsourced to private vendors; and a growing but still comparatively small presence of modern headless and no-code platforms relative to more mature markets.

Table 4 below consolidates the sector analysis presented in this chapter, summarising the typical user base, hosting model, and core mechanism associated with each major category of platform observed in Nepal.

System	Typical Users in Nepal	Hosting Model	Core Mechanism
WordPress	Small businesses, blogs, many news portals	Self-hosted (shared/VPS hosting)	Open-source, plugin-extensible
Blogger	Individual bloggers, small organisations	Google-hosted (free)	Simple, template-based publishing
Custom / in-house CMS	Large media houses, major e-commerce platforms	Self-hosted or cloud infrastructure	Purpose-built for scale and local integrations
Drupal	Institutional and NGO websites	Self-hosted	Structured, access-controlled content
Government vendor platforms	Ministries, departments, local governments	NITC-coordinated or vendor-hosted	Varies by vendor; often open-source based

Table 4: Summary of Major CMS Platforms Used in Nepal

5.6 Sector Synthesis and Comparative Observations

Viewed across sectors, Nepal's CMS ecosystem can be understood as three overlapping tiers operating on different technical assumptions. The first tier, comprising the great majority of websites by count, consists of small-business, personal, and independent-media sites built on WordPress or Blogger, prioritising low upfront cost and ease of publishing over performance headroom or multi-channel reach. The second tier, comprising a much smaller number of organisations but a disproportionately large share of total domestic web traffic, consists of Nepal's largest news portals and e-commerce platforms, which have progressively moved toward custom-built or CDN-backed infrastructure as their scale outgrew what shared or mid-tier hosting could reliably support. The third tier, government web infrastructure, sits awkwardly between the two, combining the centralised aspirations of NITC with the practical reality of a fragmented, vendor-driven procurement history that has so far prevented it from achieving either the simplicity of the first tier or the technical sophistication of the second.

A further cross-cutting observation is that Nepal-specific customisation, Bikram Sambat date handling, Devanagari typography, and integration with domestic payment and logistics systems, is most mature and most widely available within the WordPress ecosystem, reflecting the sheer size of its user base and the corresponding incentive for local developers to build and share such tools, and least mature within the SaaS-builder and headless segments, where the smaller local user base has not yet generated an equivalent volume of community-contributed, Nepal-specific extensions. This asymmetry is itself a structural factor reinforcing WordPress's dominance among Nepali organisations that require these localisation features, independent of WordPress's other general advantages of cost and familiarity.

CHAPTER 6: COMPARATIVE ANALYSIS

CMS platform categories differ across several dimensions, and the right choice depends heavily on an organisation's technical capacity, budget, and content-delivery needs. This chapter presents a structured comparison across five criteria, cost, ease of use, customisability, security, and performance, supported by an indicative total-cost-of-ownership analysis relevant to Nepali organisations planning a multi-year budget.

6.1 Total Cost of Ownership (1–5 Years)

Open-source self-hosted platforms such as WordPress carry no licensing fee, but total cost depends on hosting, premium themes or plugins, and any developer time required for setup and ongoing maintenance. SaaS website builders bundle hosting and support into a predictable monthly subscription, which can be more expensive cumulatively over several years but removes the need for separate hosting arrangements and dedicated technical staff. Headless CMS platforms often offer a generous free tier for smaller projects but require investment in custom front-end development, which is typically the largest single cost component and one that is easy for an inexperienced buyer to underestimate.

Cost Component	Open-Source Self-Hosted	SaaS Website Builder	Headless CMS
Year 1 (setup + first-year running cost, indicative)	Low – Moderate	Low – Moderate	Moderate – High
Years 2–5 (cumulative ongoing cost)	Low, rising with maintenance neglect	Moderate, predictable subscription	Moderate, dependent on usage and front-end upkeep
Dominant cost driver	Developer time for updates/security	Recurring subscription fee	Front-end development and hosting
Risk of unplanned cost	Security incident remediation	Forced plan upgrade as traffic grows	Front-end redevelopment if requirements change

Table 7: Total Cost of Ownership Comparison (1–5 Years)

6.2 Ease of Use and Skill Requirement

SaaS website builders are designed for users with no coding background and can be used to publish a functional website within hours of signing up. WordPress requires a somewhat greater initial

learning curve, particularly around plugin selection, theme configuration, and basic security hygiene, but remains accessible to non-technical users for day-to-day content publishing once a site has been set up by a developer or knowledgeable owner. Headless CMS platforms are aimed primarily at organisations with access to software-development resources, since the platform itself does not render a finished website; content editors interact with a structured content-entry interface, but a developer is required to build and maintain the consuming front-end application.

6.3 Customisability and Extensibility

Open-source platforms and headless CMS systems offer the greatest flexibility, since their code can be modified directly and extended through plugins, modules, or custom applications built against an open API. SaaS website builders trade some of this flexibility for simplicity, generally restricting users to the design and functionality options built into the platform's own app marketplace and theme system, which is a deliberate design choice that keeps the platform simple and reliable for its target audience of non-technical users.

6.4 Security and Maintenance

Self-hosted open-source platforms place the responsibility for applying security updates, managing backups, and monitoring for vulnerabilities on the site owner or their developer, which can be a significant and easily neglected burden for organisations without dedicated technical staff, a pattern commonly observed among smaller Nepali business and government websites that are set up once and rarely revisited. SaaS platforms handle these responsibilities centrally on behalf of all their customers, generally resulting in more consistent baseline security for less technically resourced users, since the vendor has both the incentive and the specialised staff to maintain platform-wide security at a level an individual small business typically could not replicate on its own.

6.5 Performance and Scalability

Performance under high traffic depends heavily on hosting infrastructure and configuration rather than the CMS category alone, though headless and custom-built architectures are generally better suited to handling very large traffic volumes and multi-channel delivery, which is why Nepal's largest news portals and e-commerce platforms tend to move toward these approaches as they scale

beyond what a standard shared-hosting WordPress deployment can comfortably support. The table below consolidates the five-criteria comparison into a single reference table.

Criteria	Open-Source Self-Hosted (e.g., WordPress)	SaaS Website Builder (e.g., Wix, Shopify)	Headless CMS (e.g., Contentful)
Upfront cost	Free software; hosting cost only	Included in subscription	Free tier; usage-based beyond limits
Technical skill required	Low to moderate	Very low (no-code)	High (requires custom front end)
Customisability	Very high, via plugins and code	Limited to platform's design tools	Very high, but requires development
Maintenance responsibility	Site owner or hired developer	Handled by the platform provider	Content team plus a development team
Multi-channel publishing	Limited without additional tools	Generally single-website only	Native strength of the architecture

Table 8: Comparative Analysis Across Five Evaluation Criteria

6.6 Illustrative Scenario Comparison

To make the preceding criteria concrete, it is useful to consider three illustrative scenarios drawn from the Nepali context described in Chapter 5. A small retail business in Kathmandu with a modest product catalogue and no in-house technical staff is generally best served by WordPress with WooCommerce on managed shared hosting, or alternatively by a SaaS builder such as Shopify if the owner is prepared to accept a recurring subscription in exchange for eliminating maintenance responsibility entirely; the illustrative scenario comparison in Table 8 suggests that the deciding factor in such cases is usually the availability of a trusted local developer relationship for ongoing WordPress maintenance, rather than any inherent technical superiority of one option over the other.

A regional news portal experiencing rapid readership growth, by contrast, is likely to reach the practical limits of a standard WordPress deployment as concurrent traffic during major news events begins to exceed what shared or entry-level VPS hosting can serve reliably; for such an organisation, the comparative analysis points toward either upgrading to more robust, cache-optimised hosting while remaining on WordPress, or beginning a phased migration toward a decoupled or headless architecture that can better absorb traffic spikes and support simultaneous publishing to a mobile application. A national-level e-commerce platform integrating multiple

domestic payment gateways and a complex, partly manual logistics workflow, finally, is generally best served by a custom-built back end from the outset, since the specific integration requirements described in Section 5.3 are unlikely to be met adequately by either a generic WordPress/WooCommerce deployment or an international SaaS commerce platform without extensive and continually maintained custom development in either case.

CHAPTER 7: RISKS, SECURITY, AND OPERATIONAL CHALLENGES

7.1 Global Supply-Chain and Plugin Risks

The open-source plugin ecosystems that make platforms such as WordPress so extensible are also a leading source of security risk. Because a typical WordPress site may rely on a dozen or more third-party plugins, each maintained by a different developer or team with differing levels of ongoing commitment, a vulnerability disclosed in any single widely used plugin can affect a very large number of otherwise unrelated websites simultaneously, a pattern sometimes described as a software supply-chain risk. Abandoned plugins, which stop receiving security updates but remain active on live sites, present a particularly persistent risk since site owners frequently do not notice that a plugin has been abandoned until a vulnerability is actively exploited.

- Frequent security vulnerabilities disclosed in popular open-source platforms and their third-party plugin ecosystems.
- Vendor lock-in risk with proprietary SaaS platforms, which can make migrating a website to another system difficult and costly.
- Rising subscription costs for SaaS and headless platforms as a website's traffic or feature needs grow beyond an entry-level plan.
- The additional development overhead required to build and maintain custom front ends for headless architectures.
- Plugin and theme compatibility issues that can arise when updating open-source CMS core software.

7.2 Local Hosting Latency and Maintenance Debt

In the Nepali context, these global risks are compounded by several local factors. Heavy reliance on a small number of platforms, chiefly WordPress, creates a concentration risk: a widely exploited vulnerability in a popular plugin could simultaneously compromise a disproportionately large share of Nepali small-business and news websites given how concentrated the local ecosystem is

around this single platform. Uneven technical maintenance among small-business and government websites, many of which are set up once by a freelancer or agency and rarely updated thereafter, produces an accumulating “maintenance debt” in which outdated CMS core software, themes, and plugins persist for years without patching. Limited local availability of specialised skills in headless and modern JavaScript-based CMS architectures, compared to more mature markets, constrains how quickly Nepali organisations can adopt these newer approaches even where the business case exists. Dependence on international hosting providers for many websites can affect load times and cost given Nepal's international bandwidth constraints, particularly for domestic visitors accessing a site hosted on a server physically located outside South Asia. Finally, inconsistent standards across government websites, reflecting the historically distributed procurement of government web platforms described in Chapter 5, produces a public-sector web presence whose security posture varies considerably from one ministry or department to the next.

7.3 Risk and Mitigation Matrix

Risk Category	Relevant To	Typical Mitigation
Plugin and software vulnerabilities	Open-source self-hosted (WordPress)	Regular updates, minimal plugin use, web-application firewalls
Vendor lock-in	SaaS website builders	Careful platform selection, regular content backups/exports
Front-end development overhead	Headless CMS	Phased adoption, use of established front-end frameworks
Inconsistent maintenance	Nepal's small-business and government sites	Managed hosting plans, centralised government IT support
Concentration risk	Nepal's CMS ecosystem as a whole	Diversification of platforms for critical, high-traffic sites
International hosting latency	Sites hosted outside South Asia	CDN adoption, evaluation of regional/local hosting options

Table 9: Risk and Mitigation Matrix

7.4 Case Illustration: Plugin Vulnerability Cascades

A useful way to understand the concentration risk described above is to consider how a single disclosed vulnerability can cascade across a large population of otherwise unrelated websites. When a security researcher or the plugin's own maintainers disclose a vulnerability in a widely installed WordPress plugin, for example, a flaw permitting unauthenticated file upload or database

access, the disclosure itself is typically accompanied by a patched plugin version. However, because WordPress plugin updates are not always applied automatically, and because many Nepali small-business and government sites are, as noted in Section 7.2, set up once and rarely revisited, a meaningful window opens during which automated scanning tools operated by malicious actors can identify and compromise every unpatched installation of that plugin globally, including in Nepal, often within days of public disclosure.

This dynamic has two practical implications for Nepali site owners and the developers who serve them. First, the security value of a plugin update is time-sensitive: applying a patch within days of release captures most of its protective value, while applying the same patch months later, after automated exploitation attempts have already had time to succeed, provides much less benefit. Second, because the vulnerability lies in the plugin rather than in WordPress core, sites that minimise their number of active plugins and remove unused ones are structurally less exposed to this category of risk regardless of how disciplined their update practices otherwise are, a straightforward, low-cost mitigation that is nonetheless inconsistently applied across the small-business and government segments of Nepal's web ecosystem.

CHAPTER 8: EMERGING TRENDS AND FUTURE OUTLOOK

Several trends are likely to shape the future of CMS adoption both globally and in Nepal. Headless and API-first architectures continue to gain adoption among organisations that need to publish the same content across a website, a mobile application, and other digital channels simultaneously, and this trend is likely to extend gradually to Nepal's larger media and e-commerce operators as technical capacity grows and as reference implementations from early domestic adopters become more widely known within the local developer community.

Headless commerce, the application of headless architecture specifically to online retail, separating the storefront presentation layer from the underlying commerce engine, is an increasingly important sub-trend internationally, allowing retailers to present a consistent product catalogue and checkout experience across a website, a mobile app, and even in-store kiosks. For Nepal's largest online retailers, headless commerce offers a plausible path toward better integration of the domestic payment and logistics workflows discussed in Chapter 5 with a more modern, maintainable technical architecture, though adoption is likely to remain limited to a small number of well-resourced players in the near term.

Edge computing, running code and caching content on servers physically distributed close to end users rather than in a single centralised data centre, is becoming more accessible to organisations of all sizes through CDN providers that now offer edge-function capabilities alongside simple content caching. For Nepal, where international bandwidth and latency remain a meaningful constraint on user experience, wider adoption of edge caching and, eventually, local or regional edge points of presence could measurably improve load times for domestically hosted content without requiring organisations to change their underlying CMS.

Artificial-intelligence-assisted content tools, including automated writing assistance, image generation, and content translation, are increasingly being integrated directly into mainstream CMS platforms such as WordPress, which is likely to lower the technical and language barriers for smaller Nepali publishers producing multilingual content in Nepali and English, and potentially in additional regional languages, without needing to hire dedicated translation staff.

In Nepal specifically, continued growth of local web-hosting providers, expanding broadband and mobile-data access, and a growing pool of locally trained developers suggest that the gap between small-site CMS practices and the more sophisticated custom and headless approaches used by larger organisations may narrow gradually over time. The gradual expansion of local and regional data-centre capacity, including growing interest from domestic hosting providers in offering higher-tier VPS and cloud products, could further reduce the international-latency disadvantage currently faced by many Nepali-hosted websites. Government efforts to modernise and standardise digital infrastructure, if extended more consistently to public-facing websites rather than focused primarily on internal systems, could likewise improve the technical consistency of government web platforms discussed in Chapter 5.

Additional developments worth monitoring include the continued growth of no-code and low-code website builders as an on-ramp for small businesses that might otherwise never have established a web presence at all; the expansion of local payment-gateway integrations directly into mainstream CMS e-commerce plugins, reducing the current reliance on bespoke agency-built integrations; and gradual experimentation with headless commerce architectures among Nepal's larger online retailers as their technical teams mature and as more locally relevant reference implementations and case studies become available to guide such investments.

CHAPTER 9: CONCLUSION AND STRATEGIC SUMMARY

CMS platforms serve distinct but complementary roles within the broader web ecosystem. Open-source, self-hosted systems such as WordPress provide a low-cost, flexible foundation that has become the default choice for the great majority of small websites in Nepal, as it has in most of the world, owing to its combination of zero licensing cost, a mature plugin ecosystem, and a large local pool of developers already fluent in the platform. Proprietary SaaS website builders and headless, API-first platforms serve more specialised needs, from no-code simplicity for organisations with no technical staff at all, to large-scale, multi-channel content delivery for organisations whose content must reach many audiences and devices simultaneously.

Nepal's CMS ecosystem reflects this global pattern while showing distinct local characteristics: a very strong concentration around WordPress for small sites, reinforced by a mature local developer community and a growing library of Nepal-specific plugins for Bikram Sambat dates and Devanagari typography; a shift toward custom and headless systems among the country's largest media and e-commerce platforms as they scale and as they seek to integrate deeply with domestic payment and logistics infrastructure that generic international platforms do not natively support; and a government web presence that combines centrally coordinated infrastructure under NITC with development still substantially outsourced to private vendors of varying technical maturity.

The comparative and risk analysis presented in Chapters 6 and 7 demonstrates that no single CMS category is universally superior; rather, each category represents a different point on a spectrum trading cost and simplicity against control and multi-channel capability. Organisations that match their choice of CMS category to their actual technical capacity, budget, and content-delivery needs, rather than defaulting to whichever platform is most familiar or most heavily marketed, are likely to see the greatest long-term benefit, whether that means a simple, well-maintained WordPress site for a small business or a headless architecture for an organisation publishing across many channels. As Nepal's digital economy continues to mature, the emerging trends discussed in Chapter 8, particularly headless commerce, edge computing, and AI-assisted content tooling, are likely to gradually narrow the current gap between the country's small-site and enterprise-grade CMS practices, provided that local technical capacity and infrastructure continue to develop in parallel.

More broadly, this case study suggests that Nepal's CMS ecosystem should not be read simply as a smaller or delayed version of global trends, but as a distinct market shaped by its own cost structures, payment and logistics infrastructure, calendar and script requirements, and government procurement practices. Effective policy and business decisions in this space therefore depend on understanding these local specificities rather than applying international benchmarks or vendor recommendations without adaptation. The findings of this report are intended to support exactly that kind of locally grounded decision-making, whether the decision at hand is a single small business choosing its first website platform or a national ministry designing a multi-year digital-infrastructure strategy.

CHAPTER 10: ACTIONABLE RECOMMENDATIONS

Recommendations arising from this study are grouped below by stakeholder category, reflecting the differing resources, risk tolerance, and technical capacity of each group. Table 10 summarises the single highest-priority action for each group before the detailed recommendations that follow.

Stakeholder Category	Top Priority Recommendation
Small and Medium Enterprises	Adopt disciplined update and backup practices on WordPress, or shift to a managed SaaS builder where no technical staff are available.
Larger Enterprises and E-commerce	Invest in CDN/edge caching and document domestic payment and logistics integrations as institutional knowledge.
Government Institutions	Establish a minimum security and maintenance baseline enforced across all vendor-built ministry and department websites.
Educational and Training Bodies	Expand curriculum coverage of headless/JavaScript CMS architectures and practical CMS security hygiene.

Table 10: Recommendations by Stakeholder Category

10.1 For Small and Medium Enterprises

- Continue to favour well-supported open-source platforms such as WordPress, paired with reputable local or regional hosting and a disciplined update schedule for core software, themes, and plugins.
- Minimise the number of active plugins to reduce the software supply-chain risk described in Chapter 7, and remove or replace any plugin that has not been updated by its developer in over a year.
- Adopt automated off-site backups and a written incident-response plan so that a compromised or corrupted site can be restored quickly rather than rebuilt from scratch.
- Where technical staff are unavailable, prioritise managed hosting plans or SaaS website builders that shift security and maintenance responsibility to the provider.

10.2 For Larger Enterprises and E-commerce Platforms

- Continue to evaluate custom or headless architectures as traffic and multi-channel publishing needs grow beyond what standard CMS deployments comfortably support.

- Invest in CDN and edge-caching adoption to mitigate the international-latency disadvantage currently facing many Nepali-hosted websites, particularly for content consumed by the diaspora audience.
- Standardise and document integrations with domestic payment gateways and logistics workflows so that this institutional knowledge does not remain solely with individual developers or a single agency.
- Conduct periodic third-party security reviews of custom-built platforms, given that these systems do not benefit from the community-driven vulnerability disclosure processes that cover major open-source CMS platforms.

10.3 For Government Institutions

- Work toward greater standardisation of web platforms and hosting under central technical coordination through NITC, to reduce inconsistency across ministries and departments.
- Establish a minimum security and maintenance baseline, covering software update cadence, backup frequency, and vulnerability monitoring, applicable to all vendor-built government websites regardless of procuring office.
- Consolidate Bikram Sambat date-handling, Devanagari typography, and bilingual publishing requirements into shared, centrally maintained components that individual departments and vendors can reuse rather than reimplementing independently.

10.4 For Educational and Training Bodies

- Expand curriculum and training-programme coverage of modern headless and JavaScript-based CMS architectures, to build the technical capacity needed for Nepal's ecosystem to keep pace with global trends.
- Incorporate practical coursework on CMS security hygiene, plugin risk assessment, and backup practices alongside conventional web-development instruction, given how frequently maintenance neglect, rather than a lack of initial technical skill, is identified as a driver of security incidents.

- Encourage student projects and case studies, such as this report, that document local adaptation patterns, since this kind of applied, Nepal-specific knowledge is not well captured in international CMS documentation or vendor materials.

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